第 4 章 不定积分

主要内容

1. 基本概念　原函数与不定积分

$$\frac{d}{dx}\left[\int f(x)dx\right]=f(x),\qquad d\left[\int f(x)dx\right]=f(x)dx,$$

$$\int F'(x)dx=F(x)+C,\qquad \int dF(x)=F(x)+C.$$

2. 计算方法

$\int f(x)dx$ → 恒等变形 / 线性运算法则 / 凑微分 / 变量代换 / 分部积分 → 基本积分表

换元法则(I)-----凑微分法

设 $F'(u)=f(u)$，$u=\varphi(x)$ 连续可导，则有

$$\int g(x)dx=\int f[\varphi(x)]\ \varphi'(x)\,\mathrm{d}x$$

$$=\int f(u)\,\mathrm{d}u\ \Big|_{u=\varphi(x)}=F(u)+C\ \Big|_{u=\varphi(x)}$$

$$=F[\varphi(x)]+C$$

换元法则(II)----变量代换

又若 $\varphi'(x)\neq 0$ $G'(x)=g(x)$ 则换元法I可逆

$$\int f(u)\,\mathrm{d}u=\int f[\varphi(x)]\ \varphi'(x)\,\mathrm{d}x=\int g(x)dx$$

$$=G(x)+C\ \Big|_{x=\varphi^{-1}(u)}=G(\varphi^{-1}(u))+C$$

分部积分公式 $\int u v' \mathrm{d}x = u v - \int u' v \mathrm{d}x$

1. 使用原则： v易求出, $\int u' v \mathrm{d}x$易积分

2. 使用经验：“反对幂指三” , 前 u 后 v'

3. 题目类型：

分部化简； 循环解出; 递推公式

注意的问题

初等函数的原函数不一定是初等函数，因此不一定都能积出。

例如，$\int e^{-x^2}\,\mathrm{d}x$， $\int \frac{\sin x}{x}\,\mathrm{d}x$， $\int \sin x^2\,\mathrm{d}x$，

$\int \frac{1}{\ln x}\,\mathrm{d}x$， $\int \frac{\mathrm{d}x}{\sqrt{1+x^4}}$， $\int \sqrt{1+x^3}\,\mathrm{d}x$，

$\int \sqrt{1-k^2\sin^2 x}\,\mathrm{d}x \ (0<k<1)$，……

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$\int \sqrt{1-k^2\sin^2 x}\,\mathrm{d}x \ (0<k<1)$，……

举　例

例1　1)求证:$\int (f(x)+f'(x))e^x dx = e^x f(x) + c$

2)设$f'(\sin^2 x) = \cos^2 x$　求$f(x)$

解1)　法1.$\int [f(x)+f'(x)]e^x dx$

$$= \int f(x)e^x dx + \int e^x f'(x)dx$$

$$= \int f(x)e^x dx + e^x f(x) - \int e^x f(x)dx$$

$$= e^x f(x) + c$$

法2.$\int\left[f(x)+f'(x)\right]e^{x}dx=\int d\left(f(x)e^{x}\right)$

$$=f(x)e^{x}+c$$

法3.令$F(x)=e^{x}f(x)+c$

$$F'(x)=e^{x}f(x)+e^{x}f'(x)$$

所以得证 ！

解2) 法1. $\because\ f'(\sin^2 x)=1-\sin^2 x$　即$f'(u)=1-u$

$$f(u)=\int(1-u)du=u-\frac{u^2}{2}+c$$

$$\therefore\ f(x)=x-\frac{x^2}{2}+c$$

法2　$f'(\sin^2 x)2\sin x\cos x=\cos^2 x\cdot 2\sin x\cos x$

$$\left[f(\sin^2 x)\right]'=2\cos^3 x\sin x$$

$$f(\sin^2 x)=\int 2\cos^3 x\sin x dx=-\frac{1}{2}\cos^4 x+c$$

$$f(\sin^2 x)=-\frac{1}{2}(1-\sin^2 x)^2+c$$

$$f(x)=-\frac{1}{2}(1-x)^2+c$$

例2 求 $\int |x-1| \mathrm{d}x$.

解 设 $F'(x)=|x-1|=\begin{cases} x-1, & x\geq 1 \\ 1-x, & x<1 \end{cases}$

则 $F(x)=\begin{cases} \frac{1}{2}x^2-x+C_1, & x\geq 1 \\ x-\frac{1}{2}x^2+C_2, & x<1 \end{cases}$

因 $F(x)$ 连续，利用 $F(1^+)=F(1^-)=F(1)$，得

$$-\frac{1}{2}+C_1=\frac{1}{2}+C_2 \overset{\text{记作}}{=\!=\!=} C$$

得 $\int |x-1| \mathrm{d}x = F(x)=\begin{cases} \frac{1}{2}(x-1)^2+C, & x\geq 1 \\ -\frac{1}{2}(x-1)^2+C, & x<1 \end{cases}$

例3 求 $\int \frac{3\cos x - \sin x}{\cos x + \sin x} \mathrm{d}x$.

解 令 $3\cos x - \sin x$

$$= A(\cos x + \sin x) + B(\cos x + \sin x)'$$

令 $a\cos x + b\sin x$

$$= A(c\cos x + d\sin x) + B(c\cos x + d\sin x)'$$

$\therefore$ 原式 $= \int \mathrm{d}x + 2\int \frac{\mathrm{d}(\cos x + \sin x)}{\cos x + \sin x}$

$$= x + 2\ln\left|\cos x + \sin x\right| + C$$

说明：此技巧适用于形为 $\int \frac{a\cos x + b\sin x}{c\cos x + d\sin x} \mathrm{d}x$ 的积分.

例4 求 $I=\int\frac{\mathrm{d}\,x}{\sin(x+a)\cdot\sin(x+b)}\quad(a-b\neq k\pi)$

解 $I=\frac{1}{\sin(a-b)}\int\frac{\sin[(x+a)-(x+b)]}{\sin(x+a)\cdot\sin(x+b)}\mathrm{d}\,x$

$$=\frac{1}{\sin(a-b)}\int\frac{\sin(x+a)\cos(x+b)-\cos(x+a)\sin(x+b)}{\sin(x+a)\cdot\sin(x+b)}\mathrm{d}\,x$$

$$=\frac{1}{\sin(a-b)}\left[\int\frac{\cos(x+b)}{\sin(x+b)}\mathrm{d}\,x-\int\frac{\cos(x+a)}{\sin(x+a)}\mathrm{d}x\right]$$

$$=\frac{1}{\sin(a-b)}\left[\ln\left|\sin(x+b)\right|-\ln\left|\sin(x+a)\right|\right]+C$$

$$=\frac{1}{\sin(a-b)}\ln\left|\frac{\sin(x+b)}{\sin(x+a)}\right|+C$$

例5 求 $I_1=\int\frac{\sin x}{a\cos x+b\sin x}\mathrm{d}x$ 及

$$I_2=\int\frac{\cos x}{a\cos x+b\sin x}\mathrm{d}x.$$

解 联合法

因为 $$aI_2+bI_1=\int\frac{a\cos x+b\sin x}{a\cos x+b\sin x}\mathrm{d}x=x+C_1$$

$$bI_2-aI_1=\int\frac{b\cos x-a\sin x}{a\cos x+b\sin x}\mathrm{d}x$$

$$=\ln\left|a\cos x+b\sin x\right|+C_2$$

$$\Longrightarrow\begin{cases}I_1=\dfrac{1}{a^2+b^2}(bx-a\ln\left|a\cos x+b\sin x\right|)+C\\[2ex] I_2=\dfrac{1}{a^2+b^2}(ax+b\ln\left|a\cos x+b\sin x\right|)+C\end{cases}$$

例6 求 $I=\int \frac{dx}{\sqrt{(x-a)(b-x)}} \quad (b>a>0)$

解法1 注意到$\frac{x-a}{b-a}+\frac{b-x}{b-a}=1$

$\therefore$ 令$\frac{x-a}{b-a}=\sin^2 t, \frac{b-x}{b-a}=\cos^2 t,$

$\Rightarrow dx=2(b-a)\sin t\cos t dt$

$$\int \frac{dx}{\sqrt{(x-a)(b-x)}}$$

$$=\int 2dt=2t+C=2\arctan\sqrt{\frac{x-a}{b-x}}+C$$

解法2 $$I=\int\frac{dx}{\sqrt{(x-a)(b-x)}}=\int\frac{dx}{(x-a)\sqrt{\dfrac{b-x}{x-a}}}$$

令 $$\sqrt{\frac{b-x}{x-a}}=t\Rightarrow x-a=\frac{b-a}{1+t^2}\Rightarrow dx=-\frac{2(b-a)t}{(1+t^2)^2}dt$$

$$I=\int\frac{2}{1+t^2}dt=2\arctan t+C=2\arctan\sqrt{\frac{b-x}{x-a}}+C$$

解法3 $$I=\int\frac{dx}{\sqrt{(x-a)(b-x)}}=\int\frac{1}{\sqrt{(\dfrac{a-b}{2})^2-(x-\dfrac{a+b}{2})^2}}dx$$

$$=\arcsin\frac{2}{b-a}(x-\frac{a+b}{2})+C$$

例6 求 $I=\int \frac{\mathrm{d}x}{\sqrt[n]{(x-a)^{n+1}(x-b)^{n-1}}}$ (n 为自然数)

$a \neq b$

解 $I=\int \frac{\mathrm{d}x}{(x-a)(x-b)\sqrt[n]{\frac{x-a}{x-b}}}$

$\frac{n}{t}\mathrm{d}t=\frac{1}{t^n}\frac{a-b}{(x-b)^2}\mathrm{d}x$

令 $t=\sqrt[n]{\frac{x-a}{x-b}}$

$\frac{n}{(a-b)t}\mathrm{d}t=\frac{\mathrm{d}x}{(x-a)(x-b)}$

则 $t^n=\frac{x-a}{x-b}$, $\boxed{n\,t^{n-1}\,\mathrm{d}t=\frac{a-b}{(x-b)^2}\mathrm{d}x}$

$=\frac{n}{a-b}\int\frac{\mathrm{d}t}{t^2}=\frac{n}{b-a}\frac{1}{t}+C=\frac{n}{b-a}\sqrt[n]{\frac{x-b}{x-a}}+C$

例6. 求 $\int \frac{\mathrm{d}x}{x^4+1}$

$$\boxed{\frac{1}{x^4+1}=\frac{1}{(x^2-\sqrt{2}x+1)(x^2+\sqrt{2}x+1)}}$$

解: 原式 $=\frac{1}{2}\int \frac{(x^2+1)-(x^2-1)}{x^4+1}\mathrm{d}x$

$$=\frac{1}{2}\int \frac{1+\frac{1}{x^2}}{x^2+\frac{1}{x^2}}\mathrm{d}x-\frac{1}{2}\int \frac{1-\frac{1}{x^2}}{x^2+\frac{1}{x^2}}\mathrm{d}x$$

注意本题技巧
按常规方法较繁

$$=\frac{1}{2}\int \frac{\mathrm{d}(x-\frac{1}{x})}{(x-\frac{1}{x})^2+2}-\frac{1}{2}\int \frac{\mathrm{d}(x+\frac{1}{x})}{(x+\frac{1}{x})^2-2}$$

$$=\frac{1}{2\sqrt{2}}\arctan\frac{x^2-1}{\sqrt{2}x}-\frac{1}{4\sqrt{2}}\ln\left|\frac{x^2-\sqrt{2}x+1}{x^2+\sqrt{2}x+1}\right|+C \quad (x\neq 0)$$

例7 求 $\int \frac{\mathrm{d}x}{1+e^{\frac{x}{2}}+e^{\frac{x}{3}}+e^{\frac{x}{6}}}$.

解 令 $t=e^{\frac{x}{6}}$, 则 $x=6\ln t$, $\mathrm{d}x=\frac{6}{t}\mathrm{d}t$

$$\text{原式}=6\int \frac{\mathrm{d}t}{(1+t^3+t^2+t)t}=6\int \frac{\mathrm{d}\ t}{(t+1)(t^2+1)t}$$

$$=\int\left(\frac{6}{t}-\frac{3}{t+1}-\frac{3t+3}{t^2+1}\right)\mathrm{d}t$$

$$=6\ln|t|-3\ln|t+1|-\frac{3}{2}\ln(t^2+1)-3\arctan t+C$$

$$=x-3\ln(e^{\frac{x}{6}}+1)-\frac{3}{2}\ln(e^{\frac{x}{3}}+1)-3\arctan e^{\frac{x}{6}}+C$$

例8 求 $\int \frac{x^2}{\sqrt{5+2x+x^2}}dx$

解法1 恒等变形, 代基本公式

$$\int \frac{x^2}{\sqrt{5+2x+x^2}}dx = \int \frac{x^2+2x+5-(2x+5)}{\sqrt{(x+1)^2+2^2}}dx$$

$$= \int \sqrt{(x+1)^2+2^2}\,dx$$

$$-\int \frac{2x+2}{\sqrt{5+2x+x^2}} - \int \frac{3}{\sqrt{(x+1)^2+2^2}}dx$$

$$= \frac{(x+1)\sqrt{5+2x+x^2}}{2} + \frac{2^2}{2}\ln(x+1+\sqrt{5+2x+x^2})$$

$$-2\sqrt{5+2x+x^2} - 3\ln(x+1+\sqrt{5+2x+x^2}) + C$$

例8 求 $\int \frac{x^2}{\sqrt{5+2x+x^2}}dx$

解法2 用三角代换 令 $x+1=2\tan t$

$$\int \frac{x^2}{\sqrt{5+2x+x^2}}dx = \int \frac{x^2}{\sqrt{(x+1)^2+2^2}}dx$$

$$= \int \frac{(2\tan t-1)^2}{2\sec t}\cdot 2\sec^2 t dt$$

$$= \int (4\sec^3 t - 4\sec^2 t\sin t - 3\sec t)dt$$

$$= \cdots$$

例8 求 $\displaystyle\int \frac{x^2}{\sqrt{5+2x+x^2}}dx$

解法3 用Euler第1代换

令 $\sqrt{5+2x+x^2} = t - x$

$$\Rightarrow x = \frac{t^2-5}{2(t+1)} \Rightarrow dx = \frac{t^2+2t+5}{2(t+1)^2}dt$$

$$\sqrt{5+2x+x^2} = t - \frac{t^2-5}{2(t+1)} — = \frac{t^2+2t+5}{2(t+1)}$$

$$\int \frac{x^2}{\sqrt{5+2x+x^2}}dx = \int \frac{(t^2-5)^2}{4(t+1)^2} \cdot \frac{1}{(t+1)}dt$$

将$(t^2-5)^2$在$t=-1$处展成Taylor公式：

$$(t^2-5)^2=16+16(t+1)-4(t+1)^2-4(t+1)^3+(t+1)^4$$

$$\int\frac{x^2}{\sqrt{5+2x+x^2}}dx=\int\frac{(t^2-5)^2}{4(t+1)^2}\cdot\frac{1}{(t+1)}dt$$

$$=\int[\frac{4}{(t+1)^3}+\frac{4}{(t+1)^2}-\frac{1}{t+1}+\frac{t-3}{4}]dt$$

$$=-\frac{2}{(t+1)^2}-\frac{4}{t+1}-\ln|t+1|+\frac{1}{8}t^2-\frac{3}{4}t+C$$

将$t=x+\sqrt{5+2x+x^2}$代入还原！

课堂练习

求下列不定积分

1. $\int \frac{\ln x - 1}{\ln^2 x} dx$

2. $\int \frac{xe^x}{(1+x)^2} dx$

3. $\int \frac{e^{3x} + e^x}{e^{4x} - e^{2x} + 1} dx$

4. $\int \frac{dx}{x(x^{100} + 1)}$

5. $\int \frac{1}{\sin^3 x} dx$

6. $\int \frac{\cos^4 x}{\sin^3 x} dx$

7. $\int \frac{dx}{x + \sqrt{1 - x^2}}$

8. $\int \frac{dx}{(x^2+1)(x^2+x+1)}$

解 1. $\int \frac{\ln x - 1}{\ln^2 x} dx = \int d\frac{x}{\ln x} = \frac{x}{\ln x} + c$

另解 令 $\ln x = t, x = e^t, dx = e^t dt$

$$\int \frac{\ln x - 1}{\ln^2 x} dx = \int \frac{t-1}{t^2} e^t dt = \int \frac{1}{t} e^t + \int e^t d(\frac{1}{t})$$

$$= \int \frac{1}{t} e^t + \frac{1}{t} e^t - \int \frac{1}{t} e^t dt = \frac{1}{t} e^t + c$$

$$= \frac{1}{\ln x} x + c$$

另解

$$\int \frac{\ln x - 1}{\ln^2 x} dx = \int \frac{1}{\ln x} dx - \int \frac{1}{\ln^2 x} dx$$

$$\because \int \frac{1}{\ln x} dx = \frac{x}{\ln x} - \int \frac{-1}{\ln^2 x} \frac{1}{x} x dx$$

$$\therefore \int \frac{\ln x - 1}{\ln^2 x} dx = \int \frac{1}{\ln x} dx - \int \frac{1}{\ln^2 x} dx$$

$$= \frac{x}{\ln x} + c$$

解　2. $\int \frac{xe^x}{(1+x)^2}dx$

法1　$\int \frac{xe^x}{(1+x)^2}dx = \int xe^x d\left(\frac{-1}{1+x}\right)$

$$= -\frac{1}{1+x}xe^x + \int \frac{1}{1+x}(1+x)e^x dx$$

法2 $$\int \frac{xe^x}{(1+x)^2}dx = \int \frac{(x+1-1)e^x}{(1+x)^2}dx$$

$$= \int \frac{1}{1+x}e^x dx - \int \frac{e^x dx}{(1+x)^2}$$

$$= \int \frac{e^x}{1+x}dx + \int e^x d\frac{1}{1+x}$$

$$= \cancel{\int \frac{e^x}{1+x}dx} + \frac{e^x}{1+x} - \cancel{\int \frac{e^x}{1+x}dx} + c$$

解 3. $\int \frac{e^{3x}+e^{x}}{e^{4x}-e^{2x}+1}dx$

法1 $$\int \frac{e^{3x}+e^{x}}{e^{4x}-e^{2x}+1}dx = \int \frac{e^{x}+e^{-x}}{e^{2x}-1+e^{-2x}}dx$$

$$= \int \frac{d(e^{x}-e^{-x})}{(e^{x}-e^{-x})^{2}+1} = \arctan(e^{x}-e^{-x})+c$$

法2 $$\int \frac{e^{3x}+e^{x}}{e^{4x}-e^{2x}+1}dx=\int \frac{(e^{2x}+1)de^{x}}{e^{4x}-e^{2x}+1}$$

$$\overset{令e^{x}=t}{=}\int \frac{t^{2}+1}{t^{4}-t^{2}+1}dt=\int \frac{\frac{1}{t^{2}}+1}{t^{2}-1+t^{-2}}dt$$

$$=\int \frac{d(t-\frac{1}{t})}{(t-\frac{1}{t})^{2}+1}$$

$$=\arctan(t-\frac{1}{t})+c=\arctan(e^{x}-e^{-x})+c$$

解 4. $\int \frac{dx}{x(x^{100}+1)}$

法1 $$\int \frac{dx}{x(x^{100}+1)} = \int \frac{x^{99}}{x^{100}(x^{100}+1)} dx$$

$$= \int \frac{d\,\dfrac{x^{100}}{100}}{x^{100}(x^{100}+1)} = \frac{1}{100}\left(\ln x^{100} + \ln(1+x^{100})\right) + c$$

法2 $$\int \frac{dx}{x(x^{100}+1)} = \int \left(\frac{1}{x} - \frac{x^{99}}{x^{100}+1} \right) dx$$

$$= \ln x - \frac{1}{100} \ln(1+x^{100}) + c$$

法3 $$\int\frac{dx}{x(x^{100}+1)}=\int\frac{dx}{x^{101}(x^{-100}+1)}=\int\frac{d\,\dfrac{x^{-100}}{-100}}{x^{-100}+1}$$

$$=-\frac{1}{100}\ln(1+x^{-100})+c$$

解 5. $\int \frac{1}{\sin^3 x} dx$

法1

$$\int \frac{1}{\sin^3 x} dx = -\int \csc x d \cot x$$

$$= -\csc x \cot x + \int \cot x(-\csc x \cot x) dx$$

$$= -\csc x \cot x - \int \csc x(\csc^2 x - 1) dx$$

$$= -\csc x \cot x - \int \frac{1}{\sin^3 x} dx + \int \frac{1}{\sin x} dx$$

$$\therefore \int \frac{1}{\sin^3 x} dx = -\frac{\csc x \cot x}{2} + \frac{\ln|\csc x - \cot x|}{2} + c$$

法2

$$\int \frac{1}{\sin^3 x}dx = \int \frac{\sin^2 x + \cos^2 x}{\sin^3 x}$$

$$= \int \frac{1}{\sin x}dx + \int \cot^2 x \csc x dx$$

$$= \ln\left|\csc x - \cot x\right| + \int \cot x d(-\csc x)$$

$$= \ln\left|\csc x - \cot x\right| - \cot x \csc x - \int \csc^3 x dx$$

$$\therefore \int \frac{1}{\sin^3 x}dx = -\frac{\csc x \cot x}{2} + \frac{\ln\left|\csc x - \cot x\right|}{2} + c$$

解

6. $\displaystyle\int \frac{\cos^4 x}{\sin^3 x}dx$

$$\int \frac{\cos^4 x}{\sin^3 x}dx = \int \frac{\left(1-\sin^2 x\right)^2}{\sin^3 x}dx$$

$$= \int \frac{1-2\sin^2 x+\sin^4 x}{\sin^3 x}dx$$

$$= \int \left(\frac{1}{\sin^3 x} - \frac{2}{\sin x} + \sin x\right)dx$$

$$= \left(-\frac{\csc x \cot x}{2} + \frac{\ln\left|\csc x - \cot x\right|}{2}\right)$$

$$-2\ln\left|\csc x - \cot x\right| - \cos x + c$$

解 7. $\int \frac{dx}{x+\sqrt{1-x^2}}$

$$\int \frac{dx}{x+\sqrt{1-x^2}} \overset{x=\sin t}{=} \int \frac{\cos t}{\sin t+\cos t}dt$$

$$=\frac{1}{2}\int \frac{\cos t-\sin t+\sin t+\cos t}{\sin t+\cos t}dt$$

$$=\frac{1}{2}\int\left(1+\frac{d\sin t+\cos t}{\sin t+\cos t}\right)dt$$

$$=\frac{1}{2}[t+\ln|\sin t+\cos t|]+c \overset{\text{还原}}{=} \cdots$$

解 **8.** $\int \frac{dx}{(x^2+1)(x^2+x+1)}$

$$\because \frac{1}{(x^2+1)(x^2+x+1)} = \frac{1}{x}\left(\frac{1}{(x^2+1)} - \frac{1}{(x^2+x+1)}\right)$$

$$= \left(\frac{1}{x} - \frac{x}{(x^2+1)}\right) - \left(\frac{1}{x} - \frac{x+1}{(x^2+x+1)}\right)$$

$$= -\frac{x}{(x^2+1)} + \frac{x+1}{(x^2+x+1)}$$

$$\therefore \int \frac{dx}{(x^2+1)(x^2+x+1)}$$

$$= \int \frac{-x}{(x^2+1)}dx + \int \frac{x+1}{(x^2+x+1)}dx$$

$$= -\frac{1}{2}\int \frac{d(x^2+1)}{(x^2+1)} + \frac{1}{2}\int \frac{d(x^2+x+1)}{(x^2+x+1)} + \frac{1}{2}\int \frac{dx}{(x^2+x+1)}$$

$$= -\frac{1}{2}\ln(x^2+1) + \frac{1}{2}\int \frac{d(x^2+x+1)}{(x^2+x+1)} + \frac{1}{2}\int \frac{dx}{(x+\frac{1}{2})^2+\frac{3}{4}}$$

$$= -\frac{1}{2}\ln(x^2+1) + \frac{1}{2}\ln(x^2+x+1) + \frac{1}{\sqrt{3}}\arctan\frac{2x+1}{\sqrt{3}} + c$$

EX

1. 设$f(x)$的原函数为 $\sin\sqrt{x}$, 求$\int xf''(x)dx$.

2、设 $\int xf(x)dx=\arcsin x+C$, 求$\int\frac{1}{f(x)}dx$.

3、已知$f(x)=\sin x$,求 $\int\frac{f'(x)\sqrt{arctgf(x)}}{1+f^2(x)}dx$.

4. 设$f(x^2-1)=\ln\frac{x^2}{x^2-2}$且$f[\phi(x)]=\ln x$, 求$\int\phi(x)dx$.

5. 设$f'(\ln x)=\begin{cases}1 & 0\le x\le 1\\ x & 1<x<+\infty\end{cases}$ 及$f(0)=0$,求$f(x)$.

1. $\int \frac{2^x 3^x}{9^x - 4^x} dx$ 2) $\int \frac{\sqrt{\ln(x + \sqrt{1 + x^2}) + 5}}{\sqrt{1 + x^2}} dx.$

2. 1) $\int \frac{dx}{x \ln x \ln \ln x}$ 2) $\int \frac{\ln \ln x}{x} dx$ 3) $\int \frac{1 + \sin x}{1 + \cos x} e^x dx$

3. 1) $\int \frac{x^2}{(x+1)^{10}} dx$ 2) $\int \frac{x^{11}}{(6+x^6)^3} dx$ 3) $\int \frac{x^7}{(1-x^2)^5} dx$

4\. 1) $\int \frac{1+\tan x}{\sin 2x} dx$ 2) $\int \frac{1-\tan x}{1+\tan x} dx$

3) $\int \frac{\sin x dx}{\sin^2 x + 4\cos^2 x}$

4) $\int \frac{\cos\sqrt{x} - 1}{\sqrt{x}\sin^2\sqrt{x}} dx$ 5) $\int \frac{1}{\sqrt{1+\sin x}} dx$

5\. 1) $\int \frac{\sqrt{x^2+a^2}}{x^2} dx$ 2) $\int \frac{x+1}{\sqrt{3+4x-4x^2}} dx$

3) $\int x^2\sqrt{3+4x-4x^2} dx$

解　答

1. 设$f(x)$的原函数为 $\sin\sqrt{x}$, 求$\int xf''(x)dx$.

解　$\int xf''(x)dx = \int x df'(x) = xf'(x) - \int f'(x)dx$

$= xf'(x) - f(x) + C$

$$f(x) = (\sin\sqrt{x})' = \cos\sqrt{x}\,\frac{1}{2\sqrt{x}} = \frac{1}{2\sqrt{x}}\cos\sqrt{x}$$

$$f'(x) = \frac{1}{4x}(-\sin\sqrt{x}) - \frac{1}{4x\sqrt{x}}\cos\sqrt{x}$$

2. 设 $\int xf(x)dx = \arcsin x + C$, 求$\int \frac{1}{f(x)}dx$.

解 $xf(x) = \frac{1}{\sqrt{1-x^2}} \quad \Rightarrow \frac{1}{f(x)} = x\sqrt{1-x^2}$

则$\int \frac{1}{f(x)}dx = \int x\sqrt{1-x^2}dx = \frac{-1}{2}\int \sqrt{1-x^2}d(1-x^2)$

$$= \frac{-1}{3}(1-x^2)^{\frac{3}{2}} + C.$$

3. 已知$f(x)=\sin x$,求 $\int \frac{f'(x)\sqrt{arctgf(x)}}{1+f^2(x)}dx$.

解 $\int \frac{f'(x)\sqrt{\arctan f(x)}}{1+f^2(x)}dx$

$$=\int \sqrt{\arctan f(x)}\,d\arctan f(x)$$

$$=\frac{2}{3}[\arctan f(x)]^{\frac{3}{2}}+C$$

$$=\frac{2}{3}[\arctan\sin x]^{\frac{3}{2}}+C.$$

4. 设$f(x^2-1)=\ln\dfrac{x^2}{x^2-2}$且$f[\phi(x)]=\ln x$，求$\int\phi(x)dx$.

解 $$f(x^2-1)=\ln\frac{x^2-1+1}{x^2-1-1}$$

$$\Rightarrow f[\phi(x)]=\ln\frac{\phi(x)+1}{\phi(x)-1}=\ln x$$

则$\phi(x)=\dfrac{x+1}{x-1}$,

$$\int\phi(x)dx=\int\frac{x-1+2}{x-1}dx=x+2\ln|x-1|+C.$$

5 设$f'(\ln x)=\begin{cases}1 & 0\le x\le 1\\ x & 1<x<+\infty\end{cases}$ 及$f(0)=0$,求$f(x)$.

解 $$f'(t)=\begin{cases}1 & 0\le e^t\le 1 \Rightarrow -\infty\le t\le 0\\ e^t & 1<e^t<+\infty \Rightarrow 0<t<+\infty\end{cases}$$

$$f(t)=\begin{cases}t+C_1 & -\infty\le t\le 0\\ e^t+C_2 & 0<t<+\infty\end{cases}$$

$$\lim_{t\to 0+}(e^t+C_2)=1+C_2=\lim_{t\to 0-}t+C_1=C_1$$

$$f(0)=C_1=0 \Rightarrow C_2=-1$$

$$f(x)=\begin{cases}x & -\infty\le x\le 0\\ e^x-1 & 0<x<+\infty\end{cases}$$

解 答

1. 1) $\int \frac{dx}{x\ln x \ln\ln x}$ 2) $\int \cos\ln x dx$ 3) $\int \frac{\ln\ln x}{x}dx$

2. 1) $\int \frac{dx}{1+e^x}dx$ 2) $\int \frac{dx}{\sqrt{1+e^x}}$ 3) $\int \frac{1+\sin x}{1+\cos x}e^x dx$

3. 1) $\int \frac{x^2}{(x+1)^{10}}dx$ 2) $\int \frac{x^{11}}{(6+x^6)^3}dx$ 3) $\int \frac{x^7}{(1-x^2)^5}dx$

4. 1) $\int \frac{1+\tan x}{\sin 2x} dx$ 2) $\int \frac{1-\tan x}{1+\tan x} dx$

3) $\int \frac{\sin x dx}{\sin^2 x + 4\cos^2 x}$

4) $\int \frac{\cos\sqrt{x} - 1}{\sqrt{x}\sin^2\sqrt{x}} dx$ 5) $\int \frac{1}{\sqrt{1+\sin x}} dx$

5. 1) $\int \frac{\sqrt{x^2+a^2}}{x^2} dx$ 2) $\int \frac{x+1}{\sqrt{3+4x-4x^2}} dx$

6. 已知$f(x)$的一个原函数为$(1+\sin x)\ln x$,求$\int xf'(x)dx$

解1

$$1)\int\frac{dx}{x\ln x\ln\ln x}=\int\frac{d\ln x}{\ln x\ln\ln x}$$

$$=\int\frac{d\ln\ln x}{\ln\ln x}=\ln\ln\ln x+c$$

$$2)\int\cos\ln x dx=x-\cos\ln x-\int x(-\sin\ln x)\frac{1}{x}dx$$

$$=x\cos\ln x+x\sin\ln x-\int x\cos\ln x\frac{1}{x}dx$$

$$2\int\cos\ln x dx=x\cos x\ln x+x\sin\ln x+c$$

$$\therefore\int\cos\ln x dx=\frac{x}{2}(\cos\ln x+\sin\ln x)+c$$

3) $\int \frac{\ln\ln x}{x}dx = \int \ln\ln x d\ln x$

$$\overset{令\ln x=t}{=} \int \ln t dt = t\ln t - \int t\frac{1}{t}dt$$

$$= t\ln t - t + c = \ln x \cdot \ln\ln x - \ln x + c$$

解2　1) $\int \frac{dx}{1+e^x}$

法1　$$\int \frac{dx}{1+e^x} = \int \frac{e^x dx}{e^x(1+e^x)} = \int \left(\frac{1}{e^x} - \frac{1}{1+e^x} \right) de^x$$

$$= x - \ln(1+e^x) + c$$

法2　$$\int \frac{1}{1+e^x} dx = \int \frac{e^x dx}{e^{2x} + e^x + (\frac{1}{2})^2 - \frac{1}{4}}$$

$$= \int \frac{de^x}{(e^x + \frac{1}{2})^2 - \frac{1}{4}} = \ln \left| \frac{e^x + \frac{1}{2} - \frac{1}{2}}{e^x - \frac{1}{2} + \frac{1}{2}} \right| + c$$

$$= x - \ln(e^x + 1) + c$$

法3 $\displaystyle\int\frac{dx}{1+e^{x}}=\int\frac{dx}{e^{x}(1+e^{-x})}=\int\frac{e^{-x}dx}{1+e^{-x}}$

$$=\int\frac{-de^{-x}}{1+e^{-x}}=-\ln(e^{-x}+1)+c$$

法4 $\displaystyle\int\frac{dx}{1+e^{x}}\overset{e^{x}+1=t}{\underset{x=\ln(t-1)}{=}}\int\frac{dt}{t(t-1)}=\int\left(\frac{1}{t-1}-\frac{1}{t}\right)dt$

$$=\ln(t-1)-\ln|t|+c\overset{\text{"还原"}}{=}x-\ln(1+e^{x})+c$$

2) 令$e^x+1=t^2$,则$x=\ln(t^2-1)$,

$$dx=\frac{t}{t^2-1}dt$$

$$\therefore \int \frac{1}{\sqrt{1+e^x}}dx=\int \frac{2t}{t(t^2-1)}dt$$

$$=2\int \frac{t}{t^2-1}dt=2\ln\left|\frac{t-1}{t+1}\right|+c$$

$$=2\ln\left|\frac{\sqrt{1+e^x}-1}{\sqrt{1+e^x}+1}\right|+c$$

$$3)\quad \int \frac{1+\sin x}{1+\cos x} e^x dx = \int \frac{1}{1+\cos x} e^x dx + \int \frac{\sin x}{1+\cos x} e^x dx$$

$$= \int \frac{1}{2\cos^2 \frac{x}{2}} e^x dx + \int \frac{2\sin \frac{x}{2} \cos \frac{x}{2}}{2\cos^2 \frac{x}{2}} e^x dx$$

$$= \int \frac{1}{2\cos^2 \frac{x}{2}} e^x dx + \int \tan \frac{x}{2} e^x dx$$

$$= \int e^x d \tan \frac{x}{2} + \int \tan \frac{x}{2} e^x dx$$

$$= e^x \tan \frac{x}{2} - \cancel{\int \tan \frac{x}{2} e^x dx} + \cancel{\int \tan \frac{x}{2} e^x dx} + c$$

解3　1)　$$\int\frac{x^2}{(x+1)^{10}}dx$$

法1

$$\int\frac{x^2-1+1}{(x+1)^{10}}dx=\int\left(\frac{x-1}{(x+1)^9}+\frac{1}{(x+1)^{10}}\right)dx$$

$$=\int\left(\frac{x+1-2}{(x+1)^9}+\frac{1}{(x+1)^{10}}\right)dx$$

$$=\int\left(\frac{1}{(x+1)^8}-\frac{2}{(x+1)^9}+\frac{1}{(x+1)^{10}}\right)dx$$

$$=-\frac{1}{7}\frac{1}{(x+1)^7}+\frac{1}{4}\frac{1}{(x+1)^8}-\frac{1}{9}\frac{1}{(x+1)^9}+c$$

法2 $\int \frac{x^2}{(x+1)^{10}}dx \overset{令x+1=t}{=} \int \frac{(t-1)^2}{t^{10}}dt$

$$= \int \frac{t^2-2t+1}{t^{10}}dt = \int (\frac{1}{t^8} - \frac{1}{t^9} + \frac{1}{t^{10}})dt$$

$$= -\frac{1}{7}\frac{1}{(x+1)^7} + \frac{1}{4}\frac{1}{(x+1)^8} - \frac{1}{9}\frac{1}{(x+1)^9} + c$$

$$2)\quad \int \frac{x^{11}}{(6+x^6)^3}dx = \int \frac{x^6 d\frac{x^6}{6}}{(6+x^6)^3}$$

$$\overset{x^6=t}{=} \int \frac{1}{6}\frac{tdt}{(6+t)^3} = \frac{1}{6}\int \frac{t+6-6}{(6+t)^3}dt$$

$$= \frac{1}{6}\int\left(\frac{1}{(6+t)^2} - \frac{6}{(6+t)^3}\right)dt$$

$$= \frac{1}{6}\left(-\frac{1}{t+6} - \frac{6}{-2(t+6)^2}\right) + c$$

$$= \frac{1}{6}\left(-\frac{1}{x^6+6} + \frac{3}{(6+x^6)^2}\right) + c$$

3) $$\int \frac{x^7}{(1-x^2)^5}dx = \int \frac{x^6 d\,\frac{x^2}{2}}{(1-x^2)^5}$$

$$= \int \frac{x^6 d\,\frac{x^2}{2}}{(1-x^2)^5} \overset{x^2=t}{=} \int \frac{\frac{1}{2}t^3 dt}{(1-t)^5}$$

$$= \frac{1}{2}\int \frac{t^3}{(1-t)^5}dt$$

法1 $\overset{1-t=u}{=} \frac{1}{2}\int -\frac{(1-u)^3}{(u)^5}du = -\frac{1}{2}\int -\frac{1-3u+3u^2-u^3}{(u)^5}du$

$$= -\frac{1}{2}\int \left(u^{-5}-3u^{-4}+3u^{-3}-u^{-2}\right)du$$

$$= -\frac{1}{2}\left(-\frac{u^{-4}}{4}-3\frac{u^{-3}}{-3}+3\frac{u^{-2}}{-2}-\frac{u^{-1}}{-1}\right)+c$$

$$= \frac{1}{2}\left(\frac{1}{4u^4}-\frac{1}{u^3}+\frac{3}{2}\frac{1}{u^2}-\frac{1}{u}\right)+c$$

$$= \frac{1}{2}\left(\frac{1}{4(1-x^2)^4}-\frac{1}{(1-x^2)^3}+\frac{2}{3}\frac{1}{(1-x^2)^2}-\frac{1}{1-x^2}\right)+c$$

法2

$$\int \frac{x^7}{(1-x^2)^5}dx \overset{x=\sin t}{=} \int \frac{\sin^7 t}{\cos^{10} t}\cos t dt$$

$$= \int \frac{\sin^7 t}{\cos^9 t}dt = \int \frac{\sin^6 t}{\cos^9 t}d(-\cos t)$$

$$\overset{\cos t=u}{=} -\int \frac{(1-u^2)^3}{u^9}du$$

$$= \int \frac{1-3u^2+3u^4-u^6}{u^9}du = \cdots$$

法3 令$1-x^2=t, 1-t=x^2, 2xdx=-dt$

$$\int \frac{x^7}{(1-x^2)^5}dx=\int \frac{(1-t)^3}{t^5}(\frac{-1}{2})dt$$

$$=\frac{-1}{2}\int \frac{1-3t+3t^2-t^3}{t^5}dt$$

$$=-\frac{1}{2}[-\frac{1}{4}\frac{1}{t^4}+\frac{1}{t^3}-\frac{3}{2}\frac{1}{t^2}+\frac{1}{t}]+c \overset{还原}{=}$$

$$=\frac{1}{2}\left(\frac{1}{4(1-x^2)^4}-\frac{1}{(1-x^2)^3}+\frac{2}{3}\frac{1}{(1-x^2)^2}-\frac{1}{1-x^2}\right)+c$$

解4

$$1)\int\frac{1+\tan x}{\sin 2x}dx=\int\frac{dx}{\sin 2x}+\int\frac{\sin x/\cos x dx}{2\cos x\sin x}$$

$$=\frac{1}{2}\ln\left|\csc 2x-\cot 2x\right|+\frac{1}{2}\tan x+c$$

$$2)\int\frac{1-\tan x}{1+\tan x}dx=\int\frac{\cos x-\sin x}{\cos x+\sin x}dx$$

$$=\int\frac{d(\sin x+\cos x)}{\cos x+\sin x}=\ln\left|\cos x+\sin x\right|+c$$

$$3)\int\frac{\sin x}{\sin^2 x+4\cos^2 x}dx=\int\frac{d(-\cos x)}{1-\cos^2 x+4\cos^2 x}dx$$

$$=-\int\frac{d\cos x}{1+3\cos^2 x}=-\frac{1}{\sqrt{3}}\arctan(\sqrt{3}\cos x)+c$$

$$4)\int \frac{\cos\sqrt{x}-1}{\sqrt{x}\sin^2\sqrt{x}}dx$$

$$=\int \frac{\cos\sqrt{x}}{\sqrt{x}\sin^2\sqrt{x}}dx-\int \frac{1}{\sqrt{x}\sin^2\sqrt{x}}dx$$

$$=2\int \frac{\cos\sqrt{x}}{\sin^2\sqrt{x}}d\sqrt{x}-2\int \frac{1}{\sin^2\sqrt{x}}d\sqrt{x}$$

$$=-2\frac{1}{\sin\sqrt{x}}+2\cot\sqrt{x}+c$$

$$5)\int \frac{1}{\sqrt{1+\sin x}}\,dx = \int \frac{1}{\sqrt{1+\cos(x-\frac{\pi}{2})}}\,dx$$

$$= \int \frac{1}{\sqrt{2\cos^2\frac{1}{2}(x-\frac{\pi}{2})}}\,dx$$

$$= \int \frac{dx}{\sqrt{2}\cos\frac{1}{2}(x-\frac{\pi}{2})}$$

$$= \sqrt{2}\ln\left|\sec\frac{1}{2}(x-\frac{\pi}{2}) + \tan\frac{1}{2}(x-\frac{\pi}{2})\right| + c$$

另解 $\int \frac{1}{\sqrt{1+\sin x}}dx = \int \frac{dx}{\sqrt{(\cos\frac{x}{2}+\sin\frac{x}{2})^2}} = \int \frac{dx}{\cos\frac{x}{2}+\sin\frac{x}{2}}$

$$= \int \frac{\frac{\sqrt{2}}{2}dx}{\cos\frac{\pi}{4}\cos\frac{x}{2}+\sin\frac{\pi}{4}\sin\frac{x}{2}}$$

$$= \frac{\sqrt{2}}{2}\int \frac{dx}{\cos(\frac{x}{2}-\frac{\pi}{4})}$$

$$= \sqrt{2}\ln\left|\sec(\frac{x}{2}-\frac{\pi}{4})+\tan(\frac{x}{2}-\frac{\pi}{4})\right|+c$$

解5　法1 1)
$$\int \frac{\sqrt{x^2+a^2}}{x^2}dx=\int \frac{x^2+a^2}{x^2\sqrt{x^2+a^2}}dx$$
$$=\int\left(\frac{1}{\sqrt{x^2+a^2}}+\frac{a^2}{x^2\sqrt{x^2+a^2}}\right)dx$$
$$=\int \frac{1}{\sqrt{x^2+a^2}}+\int \frac{a^2 d(-\frac{1}{2x^2})}{\sqrt{1+\frac{a^2}{x^2}}}$$

法2 $$\int\frac{\sqrt{x^2+a^2}}{x^2}dx=\int\sqrt{x^2+a^2}\,d\left(\frac{-1}{x}\right)$$

$$=\frac{-1}{x}\sqrt{x^2+a^2}+\int\frac{1}{x}\frac{2x}{\sqrt{x^2+a^2}}dx$$

2) $\displaystyle\int \frac{x+1}{\sqrt{3+4x-4x^2}}dx$

$$= -\frac{1}{8}\int \frac{d(3+4x-4x^2)}{\sqrt{3+4x-4x^2}} + \frac{3}{2}\int \frac{dx}{\sqrt{4-(2x-1)^2}}$$

$$= -\frac{1}{4}\sqrt{3+4x-4x^2} + \frac{3}{4}\arcsin\frac{2x-1}{2} + c$$

6. 已知$f(x)$的一个原函数为$(1+\sin x)\ln x$，求$\int xf'(x)dx$

解 $\int xf'(x)dx = \int xdf(x)$

$= xf(x) - \int f(x)dx$

$\because$ $f(x)$的一个原函数为 $(x+\sin x)\ln x$

$\therefore \left[(1+\sin x)\ln x\right]' = \cos \ln x + \dfrac{1+\sin x}{x} = f(x)$

$\int xf'(x)dx = xf(x) - (1+\sin x)\ln x$

$= x\cos x\ln x + (1+\sin x) - (1+\sin x)\ln x + c$

$= x\cos x\ln x + (1+\sin x)[1-\ln x] + c$